

SAMA Counter Fraud

Data Quality & Fraud Analytics Report

Report Date:	2026-06-01	Prepared by:	Gayasuddin
Dataset Period:	2024-01-01 – 2024-12-30	Role Target:	Counter Fraud – Data Quality Specialist
Institution:	Saudi Digital Bank (Simulated)	Regulation:	SAMA Counter Fraud Framework

1. Executive Summary

This report presents a comprehensive fraud data quality assessment for a Saudi digital banking institution. The analysis covers 50,000 transactions across full-year 2024, applying SAMA-aligned validation, deduplication, anomaly detection, and regulatory reporting frameworks. The overall Data Quality Score is **98.1%**, with key action items identified.

Total Records	Fraud Cases	DQ Score	Unreported (SAMA)
50,000	1,750 (3.5%)	98.1%	430
		■ Acceptable	■ Action Required
Duplicates Found	Null Account Nos	Anomalies (Z>3σ)	Mule Flags
499 (1.0%)	800 (1.6%)	94	192
■ Medium Risk	■ High Risk	■ Investigate	■ Escalate

2. Data Quality Assessment

2.1 Completeness Analysis

Null/missing value audit across all 16 fields in the fraud transaction dataset:

Field	Null Count	Null %	Severity	Recommended Action
fraud_type	48,250	96.5%	INFO	Expected (non-fraud records only)
device_id	2,581	5.16%	MEDIUM	Enhance mobile event capture
ip_address	1,957	3.91%	MEDIUM	Mandatory for CNP transactions
merchant_category	1,000	2.00%	HIGH	Required for typology mapping
account_number	800	1.60%	HIGH	Critical – mandatory for SAMA reporting

2.2 Deduplication Results

Metric	Count	Rate	Status
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Total Records Ingested	50,000	100%	—
Exact Duplicate Transaction IDs	499	1.00%	■ Flagged
Near-Duplicates (Same Cust+Amt, <60s)	0	0%	■ Clean
Records After Deduplication	49,501	99.0%	■ Clean

3. Fraud Analytics & Trend Analysis

3.1 Fraud Typology Breakdown (SAMA Categories)

Fraud Typology	Cases	Total Value (SAR)	Avg Value (SAR)	Share %
Card-Not-Present (CNP)	374	1,227,721	3,283	21.4%
Account Takeover (ATO)	362	720,937	1,992	20.7%
Identity Theft	334	1,307,345	3,914	19.1%
Mule Account Activity	334	1,007,461	3,016	19.1%
Phishing	346	708,796	2,049	19.7%
TOTAL	1,750	4,972,260	2,841	100%

3.2 Anomaly Detection – Z-Score Analysis

Statistical anomaly detection was applied using Z-Score on completed transaction amounts. Transactions with $|Z| > 3\sigma$ were flagged as anomalous. **94 anomalous transactions** were identified with a combined value of **SAR 27.9M**. The fraud rate among anomalies (7.4%) is 2x the baseline rate (3.5%), validating this approach.

Metric	Value
Mean Transaction Amount	SAR 2,477.55
Standard Deviation	SAR 19,945.06
Anomaly Threshold	$ Z\text{-Score} > 3\sigma$
Anomalous Transactions	94
Fraud Among Anomalies	7 cases (7.4% — 2x baseline)
Total Anomaly Value	SAR 27,917,671.68
Recommendation	Auto-route to manual review queue

4. SAMA Regulatory Reporting

■ **REGULATORY ACTION REQUIRED**

430 confirmed fraud cases (SAR 939,088.87) have not been reported to SAMA. Per SAMA Counter Fraud Framework, these must be submitted within the **72-hour mandatory window**.

4.1 Monthly Fraud Report (SAMA Format)

Month	Total Txns	Fraud Cases	Fraud Value (SAR)	Fraud Rate	SAMA Reported
Jan 2024	4,161	147	419,077	3.53%	110 (75%)
Feb 2024	4,089	145	190,521	3.55%	109 (75%)
Mar 2024	4,151	132	786,201	3.18%	99 (75%)
Apr 2024	4,003	128	126,308	3.20%	96 (75%)
May 2024	4,195	136	180,633	3.24%	102 (75%)
Jun 2024	4,118	148	170,740	3.59%	111 (75%)
Jul 2024	4,123	139	152,096	3.37%	104 (75%)
Aug 2024	4,259	152	604,242	3.57%	114 (75%)
Sep 2024	4,130	146	532,809	3.54%	110 (75%)
Oct 2024	4,247	152	485,009	3.58%	114 (75%)
Nov 2024	4,039	147	178,156	3.64%	110 (75%)
Dec 2024	3,986	160	1,132,782	4.01%	120 (75%)
TOTAL	49,501	1,732	4,958,574	3.50%	1,299 (75%)

5. Recommendations & Action Plan

Priority	Issue	Recommended Action	Owner	Timeline
■ HIGH	SAMA Unreported Fraud	Submit 430 cases via SAMA portal immediately	Compliance/MLRO	72 hrs
■ HIGH	Missing account numbers	Add mandatory field validation at source	Data Engineering	1 week
■ MED	Duplicate tx records	Implement dedup at ETL ingestion layer	Data Management	2 weeks
■ MED	Missing IP/device data	Enforce capture in mobile SDK & API gateway	Engineering	2 weeks
■ MED	High-risk non-fraud records	Review risk scoring model threshold calibration	Fraud Analytics	1 month
■ LOW	Anomaly routing	Auto-flag Z >3σ transactions to manual review	Fraud Ops	1 month
■ LOW	Mule detection	Implement network graph analysis for linked accounts	Data Science	2 months